

A Literature-Implied Answer to Conjecture 1 of arXiv:1904.09373

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Status. Literature-implied answer (full answer to Conjecture 1 only). The implication below is a direct application of the classical measurable Turán–Nazarov inequality. It is not presented as a new theorem. The supporting authors did not know they were answering Lawton’s later question.

Source question. Wayne M. Lawton, *Distribution of Small Values of Bohr Almost Periodic Functions with Bounded Spectrum*, arXiv:1904.09373 (2019), Conjecture 1.1 on page 2 of the locally compiled source PDF, asks whether the sequence C_n in

$$J_f(u) \leq C_n H_f^{-1/n} u^{1/n}, \quad u > 0, \quad (1)$$

can be replaced by a bounded sequence. Here $f(x) = \sum_{j=0}^n a_j e^{i\omega_j x}$ has distinct real frequencies, $H_f = \max_j |a_j|$, and

$$J_f(u) = \lim_{T \rightarrow \infty} \frac{1}{2T} |\{x \in [-T, T] : |f(x)| < u\}|.$$

Supporting theorem. Friedland and Yomdin quote Nazarov’s measurable Turán–Nazarov theorem as Theorem 1.1 on page 1 of arXiv:1107.0039. In the purely imaginary exponent case needed here, it says that an absolute constant $A \geq 1$ satisfies

$$\sup_I |f| \leq \left(\frac{A|I|}{|E|} \right)^n \sup_E |f| \quad (2)$$

for every interval $I \subset \mathbb{R}$ and every measurable $E \subset I$ of positive measure. Crucially, the exponent is the number of terms minus one and A is independent of the frequencies and of n .

Theorem 1. *For every trigonometric polynomial with at most $n + 1$ distinct real frequencies and every $u > 0$,*

$$J_f(u) \leq A H_f^{-1/n} u^{1/n}.$$

Consequently Lawton’s Conjecture 1.1 is true, with the bounded sequence $C_n = A$.

Proof. The case $f = 0$ is excluded in the source statement. Write $I_T = [-T, T]$, set

$$E_T = \{x \in I_T : |f(x)| < u\}, \quad \delta_T = \frac{|E_T|}{|I_T|}, \quad M_T = \sup_{I_T} |f|.$$

If $|E_T| = 0$, then $\delta_T = 0$. Otherwise, applying (2) to $E_T \subset I_T$ and using $\sup_{E_T} |f| \leq u$ gives

$$M_T \leq \left(\frac{A}{\delta_T} \right)^n u, \quad \text{hence} \quad \delta_T \leq A \left(\frac{u}{M_T} \right)^{1/n}.$$

Choose j_0 with $|a_{j_0}| = H_f$. Orthogonality of distinct real frequencies in the Bohr mean gives

$$a_{j_0} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(x) e^{-i\omega_{j_0} x} dx.$$

The absolute value of each average on the right is at most M_T ; therefore $H_f \leq \liminf_{T \rightarrow \infty} M_T$. Since $J_f(u) = \lim_{T \rightarrow \infty} \delta_T$ exists, taking the upper limit in the preceding sublevel estimate yields

$$J_f(u) \leq A \left(\frac{u}{H_f} \right)^{1/n}.$$

This is exactly (1) with $C_n = A$ for every n . □

Identification and scope. The original 1993 Nazarov theorem predates arXiv:1904.09373, and Friedland–Yomdin’s 2011 restatement does not cite Lawton. Thus this is an agent-identified corollary rather than a later paper explicitly resolving the conjecture. No claim is made here about Conjectures 2 or 3 of Lawton’s paper.

Search evidence. The bounded search used the exact source title, the exact bounded- C_n conjecture phrase, the author name, and the terms “small values,” “Bohr almost periodic,” and “Turán–Nazarov inequality.” It found the source paper and the classical measurable-set inequality, but no separate paper explicitly recognizing this implication.

References

- [1] W. M. Lawton, *Distribution of Small Values of Bohr Almost Periodic Functions with Bounded Spectrum*, arXiv:1904.09373, 2019.
- [2] O. Friedland and Y. Yomdin, *An observation on the Turán–Nazarov inequality*, arXiv:1107.0039, 2011. See Theorem 1.1.
- [3] F. L. Nazarov, *Local estimates for exponential polynomials and their applications to inequalities of the uncertainty principle type*, Algebra i Analiz 5 (1993), no. 4, 3–66; English translation, St. Petersburg Math. J. 5 (1994), 663–717.